



Review article

A comprehensive review on deep learning approaches for short-term load forecasting

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ABSTRACT

The balance between supplied and demanded power is a crucial issue in the economic dispatching of electricity energy. With the emergence of renewable sources and data-driven approaches, demand-side or demand response (DR) programs have been applied to maintain this balance as accurately as possible. Short-term load forecasting (STLF) has a decisive impact on the success, sustainability, and performance of those programs. Forecasting customers' consumption over short or long time horizons allows distribution companies to establish new policies or modify strategies in terms of energy management, infrastructure planning, and budgeting. Deep learning (DL)-based approaches for STLF have been referenced for a long time, considering factors such as accuracy, various performance measures, volatility, and adverse effects of uncertainties in load demand. Hence, in this review, DL-based studies for the STLF problem have been considered. The studies have been classified by several titles, such as the provided method and main ideas, dataset specifications, uncertain-aware approaches, online solutions, and practical extensions to DR programs. The main contribution of this review is the ongoing exploration of STLF with DL models to reveal the research direction of the load forecasting problem in terms of the future-oriented integration of the key concepts of online, robustness, and feasibility.

1. Introduction

The pursuit of economical, efficient, reliable, and safe energy sources integrated into autonomous heterogeneous power grids has led to the development of the smart grid (SG) concept. This concept provides a versatile platform for implementing intelligent solutions from various perspectives, including human-centric approaches, technological compliance, technical harmony, and environmental friendliness. The distribution networks have become increasingly decentralised due to the rapid and growing penetration of microgrid networks powered by renewables. Moreover, the digital revolution has introduced new technologies into our daily lives, shaping and influencing our preferences and individual standards. In this context, electricity consumption data of residents has become a crucial tool for managing and regulating SG networks through demand response (DR) programs. DR refers to the strategies employed in the electricity market to manage and balance the supply and demand of electricity during different periods, including real-time [1]. Based on each consumption profile, DR includes policies to encourage consumers to voluntarily reduce or shift their electricity consumption during specified periods. However, the full deployment of data-meter devices and their maintenance and online operation can be costly or unfeasible. Accurate residential load forecasting (LF)

has, therefore, become a popular research topic for both small and large-scale DR programs. LF can be divided into two main categories based on the time horizon of the data. Short-term load forecasting (STLF) involves predicting electricity demand in the relatively short-term, typically a few minutes, hours, or days ahead, while long-term LF relates to predictions up to 20 years ahead [2]. LF minimises the risk of energy supply and demand imbalances. With the help of LF, a feasible and reliable energy management framework can be established with DR and demand-side programs. Such a decision-making framework offers advantages for both power suppliers and end-users while aligning with government commitments and regulations. Therefore, LF plays a significant role in the implementation of each micro decision-making-based component of the SG.

The primary challenge in achieving accurate LF results is the complexity of grid operations. Various factors, including user preferences, variations in the quality and quantity of appliance types, the prevalence of renewables, and fluctuating prices, contribute to this complexity. Among these factors, fluctuations in prices and supplied power from alternative sources are particularly significant. Unidirectional grids without price regulation applications lack interaction between customers and utilities, preventing end-users from reacting to price variations [3].

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Nomenclature

ANN	Artificial Neural Network
ARIMA	Autoregressive Integrated Moving Average
BMN	Boltzman Machine Network
CNN	Convolution Neural Network
DBN	Deep Belief Network
DL	Deep Learning
DR	Demand Response
GA	Genetic Algorithm
GB	Gradient Boosting
GBM	Gradient Boosting Machine
GRU	Gated Recurrent Unit
IoT	Internet of Things
k -NN	k -Nearest Neighbour
LD	Load Demand
LF	Load Forecasting
LSTM	Long-Short Term Memory
LT	Load Type
ML	Machine Learning
MLP	Multi-layer Perception
NN	Neural Network
PSO	Particle Swarm Optimisation
Ref	Reference
RES	Renewable Energy Source
RF	Random Forest
RL	Reinforcement Learning
RNN	Recurrent Neural Network
SG	Smart Grid
SR	Sample Rate
STLF	Short-Term Load Forecasting
SVM	Support Vector Machine
SVR	Support Vector Regression
TCN	Temporal Convolutional Networks

However, with tariff-based regulations and the growing preference for bidirectional renewable energy-based microgrids, end-users can now interact with energy suppliers, paving the way for dynamic pricing concepts in the energy market. This transformation has been facilitated by smart meter-based time-ahead LF or survey-based passive LF techniques.

The literature reflects a focus on improving and proposing new concepts for energy management. The outcomes of these studies can be directly linked to financial benefits for both utility companies (suppliers) and end-users in terms of reducing operation and energy costs, respectively. Deep learning (DL)-based LF techniques have gained prominence due to the constraints in electrical energy production and consumption cycles. In this context, data-driven LF solutions offer distinct advantages for managing load demand (LD) and the supply of renewable energy sources (RESs). Data-driven techniques can be applied on the supply side to integrate renewables. This integration is facilitated through the mutual interaction of customers and utilities. By analysing consumption patterns, energy sources can be installed closer to the points of consumption. The use of renewable resources can proportionally reduce the environmental impact of conventional resources. Diversifying energy sources enhances energy security by reducing exposure to fluctuations in fuel prices and supply disruptions. Therefore, machine learning (ML) and DL-based neural network (NN) approaches can be used to coordinate renewable energy production and adjust grid operations in real-time to maintain stability. On the supply side, the main concern

is that a large quantity of electrical energy must be consumed immediately after generation. Renewable-powered microgrids have mitigated this constraint to some extent. However, the demand side remains volatile and unpredictable, as evident in load profiles. The primary non-linear relationships in LD profiles can be attributed to occupancy behaviours, seasonal variables, energy costs, geographical conditions, etc. Therefore, LF models must possess self-adaptability, repeatability, and robustness to handle contingencies [4]. Given the volatile nature of residential LD, an LF model should be adaptable to preference-based trends, repeatable for forward predictions, robust against uncertainties, and validated for expected accuracy. In this regard, DL approaches offer reasonable and profitable solutions, despite the unpredictable nature of the load profiling problem, thanks to their substantial non-linear mapping capabilities. The main types of DL methods for LF problems include recurrent neural network (RNN), variants of RNN such as long short term memory (LSTM) networks or gated recurrent unit (GRU), and convolutional neural networks (CNN) [5]. In addition to these individual techniques, combinations of approaches have been proposed.

In this context, hybrid and ensemble LF models have been developed to enhance the performance of NN models. A hybrid LF model combines two or more models to solve a specific problem, leveraging the strengths of each model to produce a more accurate solution. For example, a hybrid LF model might combine a time series model, such as an autoregressive integrated moving average (ARIMA), with another NN model to capture both linear and non-linear dynamics in the data. An ensemble LF model involves training multiple models on the same data and combining their predictions through a single output to produce more robust forecasts. Ensemble models can be used to reduce the variance or bias of predictions and improve overall forecast accuracy. In the context of an LF problem, an ensemble model may involve training multiple decision trees or sub-models and combining their predictions through a voting mechanism or by averaging their predictions.

Considering the related review papers, the concepts of DL or LF can be specified with respect to the scope of the work. In this manner, DL-based studies can be reviewed with or without limitations on the referred techniques or time range. LF can be considered as a sub-item of a study on electrical power systems applications or as an independent subject within its own internal classification. Besides, LF can also be addressed by load types (LTs), forecast horizon, and time range. In this regard, the reviews can be classified into classes. These classes are the DL concept with special LTs for forecasting, DL applications to the electric power systems, deep forecasting models, and a special form of DL-based LF problem.

Considering the first class, review research is conducted by focusing on the RNNs technique as well as touching on some other ML techniques for residential-type LF [6]. In this manner, the works on DL-type LSTM, CNN, and RNNs, as well as other approaches based on ML and statistical tools, are surveyed, and their results are analysed for building LF [7]. Using the same data type, the pros and cons-based review analysis is performed [8,9].

At the second class of reviews, DL-based papers subjected to the applications of electric power systems are examined. In these reviews, DL-based LF can be a small part of the study, and LF is generally not surveyed in detail [10–13]. Another class includes the research focused on deep forecasting models. These studies consider the modelling, tuning, and applications stages of DL techniques and provide detailed investigations on the stages of determining the dynamics and parameter values within the model as well as performance indices [14–16].

The last class of the related reviews aims to search LF-based papers in terms of the forecast horizon and DL techniques. The concept of LF with a specific horizon or LF covering all horizon types is selected. Besides, single or a group of DL techniques are selected. In this regard, hybrid DL models based on nature-inspired meta-heuristic techniques for STLF are investigated [17]. Another paper deals with DL techniques

for the general LF problem [18]. Following the survey between 2015–2020, a conceptual DL learning model is proposed, which promises to reduce computational time in LF. Deep and shallow learning techniques associated with all horizon-type LF are dealt with to analyse the application suitability of the techniques [19]. Following the same learning scope, a review is performed for STLF models by [20]. This paper mainly focuses on the performance analysis of STLF techniques and includes a limited number of studies.

Based on the above analysis, it is evident that there are several research gaps in the field of STLF-DL. While the literature contains a plethora of sources, there is a scarcity of reviews tailored to meet the future needs of this field. Furthermore, the future directions outlined in existing review papers often focus on specific models and dataset specifications. In this review, we have addressed a specialised problem to anticipate the evolving demands related to expanding datasets, end-user preferences, and energy market requirements. Therefore, the customised problem of DL-based STLF has been selected as a focal point. Additionally, the future direction of this problem is designed to encompass the upcoming requirements, including online capabilities, uncertainty awareness, and practical implementability. This review also encompasses all-time papers related to DL-based STLF and offers key insights into future directions.

Power systems in almost every country have recently been privatised, and electricity is now bought and sold in the energy market [2]. In this context, the stability of energy prices can only be ensured by a realistic indicator known as STLF. Consequently, STLF has become a critical element in regulating market prices, allowing for trade-off negotiations between the increasing operational competition brought about by the SG concept and users' demand for convenient and cost-effective solutions. Despite the vast potential of STLF to meet the needs of both market parties by offering economical and comfortable solutions to the LF problem, it poses challenges due to various factors that complicate accurate electricity consumption prediction. These factors encompass temporal variability with daily, weekly, and seasonal patterns, sensitivity to weather conditions (including temperature, humidity, wind speed, and radiation levels), rapid load fluctuations resulting from real-time failures like equipment malfunctions, unplanned outages, and performance degradation due to intermittent renewable sources, price dynamics, uncertainty factors, dataset quality, modelling limitations, and the growing number and diversity of domestic loads [21].

When considering these factors, the impact of weather conditions on forecasting quality, as well as rapid load fluctuations, are noteworthy. Weather conditions significantly affect the accuracy of predictions, and rapid load fluctuations result from real-time failures and intermittent renewables. Price dynamics and uncertainty factors encompass unexpected occurrences within the systems, making them vital for informed decision-making and experience accumulation. Ensuring high-quality, well-granulated historical data for the dataset is crucial for understanding consumer behaviour. Finally, modelling limitations, while simplifying forecasting models for ease of interpretation and computation, need to be carefully considered [22]. Addressing these limitations is essential in the face of the various factors influencing electricity consumption.

Given the aforementioned challenges, effectively addressing the complexities of the STLF problem demands robust and adaptive forecasting models [23]. In this context, DL approaches present a viable solution to meet the modelling requirements of STLF. DL methods have consistently proven their mettle as robust and adaptive forecasting models, particularly when dealing with extensive time series datasets. Nevertheless, it is crucial to acknowledge that DL methodologies are not universally applicable, and their suitability should be assessed based on the specific demands of the LF problem. These considerations may encompass the availability of data, computational complexity, and interpretive aspects relevant to the application. It is often beneficial to explore a spectrum of modelling approaches, which could include traditional statistical methods and ML algorithms, when

tackling forecasting tasks. Additionally, classical metrics or customised metrics reflecting user or regional behaviours and occupancies can be readily incorporated into DL-based methods [24].

Another crucial factor to consider is the compatibility of DL-based solutions with the increasing decentralisation of distribution networks, prompted by the rapid proliferation of RESs. This matter can be assessed by examining grid monitoring and control, as well as the integration of distributed energy resources. These factors have significantly heightened the complexity of LF models. Therefore, the extensive mapping capabilities of DL-based models stand out as a prominent solution to meet the urgent requirement for rapid price determination within short-time horizons.

Last but certainly not least, it is vital to acknowledge that preferences and individual standards are continuously evolving [25,26]. The DL-based STLF concept possesses the substantial capability to incorporate new or modified cause-and-effect relationships into the model. This approach can be exemplified by focusing on factors such as future engineering, consumer segmentation, the integration of recent survey data, and the feedback mechanism.

Regarding future engineering, the model is enriched by including information on energy-efficient appliances, shifts in lighting and temperature preferences, and occupancy patterns of end-users. Furthermore, consumer segmentation enables a more personalised forecast by categorising consumers into distinct groups with unique preferences. Additionally, survey data provides valuable insights to enhance forecast accuracy by capturing recent preferences and standards. Finally, the feedback mechanism, achievable through an online-iterative based optimisation procedure allows for adjustments to the internal dynamics of the LF model. This concept is particularly relevant for the model architecture of forward-looking real-time applications that consider online and uncertainty dynamics.

Building upon the discussions in the preceding section, it is evident that achieving the required accuracy in addressing the LF problem necessitates a short-term dataset horizon and a flexible modelling approach. The model should possess a robust capability to map both predicted and random variations in energy market parameters and effectively integrate RESs. To tackle this challenge, we propose the hypothesis that “DL-based models for STLF are essential for the dynamic and feasible operation of the energy market”. Precisely predicting STLF from an energy market operator dataset is a formidable task due to the considerable uncertainty and volatility in demand, coupled with a relatively short prediction interval. ML-based models rooted in DL concepts have often been explored to address these complexities, leading to a substantial body of literature dedicated to STLF. Additionally, outlining the future directions of DL-STLF requires an innovative perspective and the synthesis of background information.

In this study, we introduce DL-based STLF models and conduct a comprehensive review. To our knowledge, prior research has not thoroughly explored the diverse facets of the STLF problem, including problem types, modelling forms, and future solutions. This review provides a detailed analysis of relevant studies, highlighting emerging trends and encompassing online DL-based STLF models, uncertainty-aware models, practical evidence, and dataset details. We propose a forward-looking concept to address future directions in online, robust, and implementable aspects.

This research paper primarily focuses on the examination of studies related to STLF problems using DL approaches, covering various aspects such as techniques proposed, key ideas in reviewed papers, dataset origins and types, uncertainty awareness, online methodologies, and practical evidence in DR applications. The paper classifies the concept into categories like online models, uncertainty-aware perspective, and practical evidence context to provide a comprehensive solution for DL-based STLF. Within the framework of the SG concept, it aims to generate precise input data for DR applications and offers references for future LF or DR studies. The ultimate goal is to provide a unified direction for future research on DL-based STLF, with key contributions

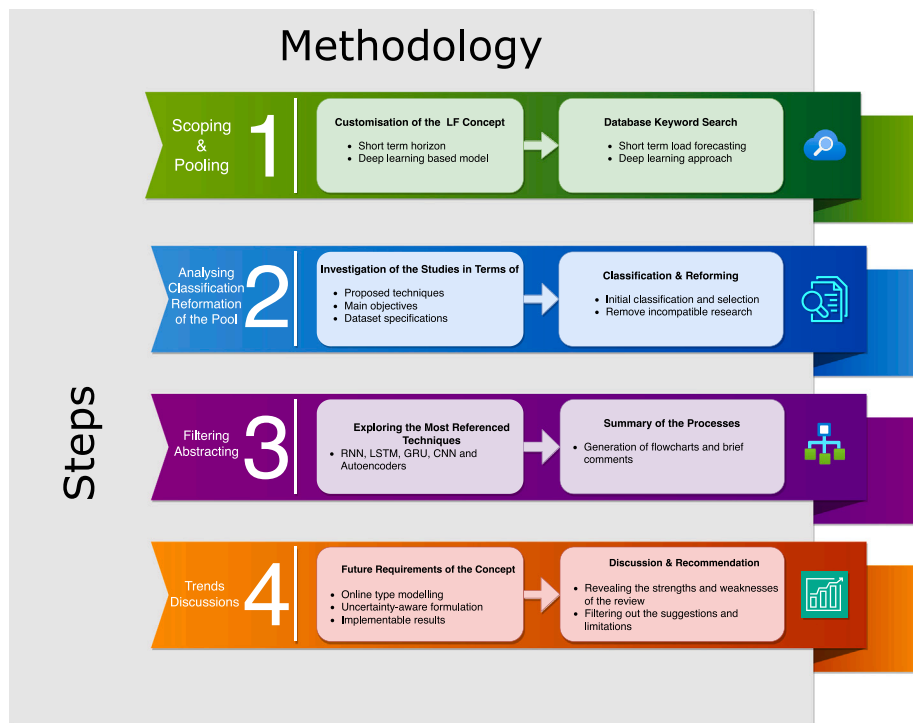


Fig. 1. Overview of the research methodology.

including a detailed analysis of relevant studies, dataset descriptions, explanations of commonly referenced techniques, and the highlighting of important concepts for future research in the field.

The rest of this review is organised as follows. Section 2 explains the methodology employed in this review. Section 3 introduces the LF problem for electricity consumption. Section 4 provides a summary of DL-based studies for STLF problems, accompanied by detailed tables. Section 5 includes concise information about frequently referenced techniques. Sections 6 and 7 delve into notable online approaches and uncertainty awareness, respectively. Section 8 scrutinises the practical evidence of DL-based STLF. Section 9 offers comprehensive discussions regarding future directions, recommendations, and limitations. Finally, Section 10 concludes the review.

2. Methodology

The methodology, illustrated in Fig. 1, comprises the following steps: Scoping and Pooling, Analysis, Classification, and Pool Refinement; Filtering and Abstraction; Identification of Future Research Trends and Discussions. These steps are elaborated as follows:

Step 1: Scoping and pooling. The specific problem definition for the LF problem, as well as the determination of the solution model type, has been carried out. The STLF concept is identified as a specialised form of the LF problem, given its potential to effectively address end-user energy usage patterns. STLF has gained significance due to the increasing number of parameters driven by higher expectations for economical and comfortable solutions. Therefore, DL-based models have emerged as the most preferred techniques, owing to their substantial capacity to map input–output relationships and make visible user behaviours related to electricity demand.

In this context, a literature survey has been conducted using the keywords “STLF” and “DL”. Studies that include both of these keywords were searched using academic search engines (Web of Science and Google Scholar). The details of this step are provided in Section 3.

Step 2: Analysing, classification and reformation of the pool. The studies have been categorised based on the proposed techniques and primary objectives, as well as dataset specifications. The details of these classifications can be found in Tables 1 and 2. The primary aim of this categorisation is to highlight the variations in requirements that can guide future research directions for the specified problem. Additionally, a separate dataset review has been conducted, considering different LTs, origins, and sample rate (SR). By simultaneously examining the contents of these tables, prospective researchers can easily establish connections between proposed techniques and objectives, facilitating the exploration of future ideas. The specifics of this step are provided in Sections 3 and 4.

Step 3: Filtering and abstracting. In light of the review, the most frequently referenced DL-based modelling techniques are identified as RNN, LSTM, GRU, CNN, and autoencoders. These techniques are briefly explained using flowcharts and their characteristic features. Further details on Step 3 can be found in Section 5.

Step 4: Identification of the future research trends and discussions. Future requirements of the DL-based STLF problem are defined by three critical aspects: online-type modelling, uncertainty awareness, and implementability. The shared perspectives on these concerns enhance the feasibility of future efforts to meet the performance demands of the modern SG concept. There is a limited body of literature that addresses the performance of individual or combined objectives within these areas. A detailed analysis of these specifics is presented in Sections 6, 7, and 8. Furthermore, a comprehensive discussion of the review, encompassing key ideas and additional perspectives, can be found in Section 9.

3. LF for electricity consumption

LF leverages local patterns in the LD of end-user groups, presenting a challenging problem due to the unpredictable behaviours of occupants and the influence of exogenous parameters on electricity consumption, such as weather conditions, local or nationwide human activities, festivals, and more. The scope of LF is categorised into four

Table 1
Summary of DL-based STLF studies: Techniques and objectives.

Ref.	Year	Proposed technique(s)	Main objective
[27]	2016	DBN model with multiple layers of restricted boltzmann machine network (BMN)	To compare and evaluate the performance of the proposed model with feedforward multi-layer perception (MLP) NN model.
[28]	2017	RNNs	To exploit the potential of DL approach in LF.
[29]	2018	Deep feedforward network and probability density forecasting methods based on quantile regression and kernel density estimation	To conduct the comparative analysis in terms of the performance evaluations of the proposed method and other competitive methods.
[30]	2018	Combination of LSTM and feed forward NN methods	To search the best learning factor in the patterns of consumption trends on the day types.
[31]	2018	Feedforward and deep RNNs	To demonstrate the applicability of deep NNs for the aim of appliance-level STLF.
[32]	2018	Deep NN based model by combination of CNNs and variation autoencoders	To search rich feature extraction case in order to build generalised appliance-level LF by proposing deep NN based energy disaggregation structure.
[33]	2018	Adapted DL algorithms of denoising autoencoder, RNN and rectangle type network	To address the high uncertainty in households' demand for STLF, a model is provided based on a feedforward ANN, along with a preprocessing stage employing energy disaggregation techniques.
[34]	2018	Deep NN with MLP	To find optimal operation procedure of the residential appliances using DL.
[35]	2018	DL-based load profile clustering framework with convolutional autoencoder	To incorporate daily and seasonal variations into the cluster framework with reduced data dimensionality.
[36]	2018	Temporal ensemble learning model with kernel regression, SVR and other NN approaches.	To compensate the lack of multivariate data in the dataset by proposing ensemble approach based on temporal features.
[37]	2019	LSTM with multi-input multi-output	To address non-linearity and other ill-defined dependencies in the data, they employ a novel prediction framework that is entirely driven by historical data, eliminating the need for information about dataset characteristics.
[38]	2019	Supervised based DL models with one-step secant backpropagation ANN and Fletcher-Broyden-Shanno-Goldfarb quasi-Newton network	To overcome the design challenges for district level electricity distribution by DL forecasting model.
[39]	2019	DL forecasting by probability density functions	To consider the influencing factors on the electric consumption in a DL forecasting model.
[3]	2019	Multiple-inputs multiple-outputs deep RNNs	To consider relationship between price and load, and proposes two DL forecast models to utilise further to the new DR programs.
[40]	2019	Ensemble model with gradient boosting (GB) regression, MLP and LSTM	To present stacked multi-learning ensemble model for near real-time residential energy demand forecasting.
[41]	2019	Probabilistic LF approach with multitask Bayesian DL	Given the uncertainty and volatile nature of residential load, they have developed a Bayesian DL model with a high level of reliability and stability.
[42]	2019	Probabilistic baseline estimation framework with DL-based optimal clustering stage	To improve the prediction process by considering the large number of daily patterns with DL-based clustering approach.
[43]	2019	CNN with recurrence plots which encode time series data into images	To propose a DL-based LF solution for DR applications targeting single residential loads, which typically exhibit higher volatility compared to aggregated loads.
[44]	2019	RNN with differentiable architecture search	To show the merit of the proposed RNN with optimised hyper parameters.
[45]	2019	Hybrid ensemble deep belief learning approach with k -NN classification method	To propose an ensemble model that leverages the high generalisation, adaptability, and non-linear mapping capabilities of DL.
[46]	2019	DBN model with Gauss-Bernoulli restricted BMN	To enhance the effectiveness of the DBN-based LF by customising the RBM technique and incorporating optimisation procedures to fine-tune the network parameters.
[47]	2019	LSTM	To improve performance and incorporate weather-based variables into the proposed LF model for assessing discomfort index.
[48]	2019	Stacked bidirectional LSTM	To incorporate meteorological data, a significant source of uncertainty, into the proposed model.
[49]	2019	Deep NN-CNN	To provide integrated approach with adequate accuracy to pre-process and analyse the consumption.

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segments: very short-term forecasting, short-term forecasting, mid-term forecasting, and long-term forecasting [51]. Each category encompasses different time spans, ranging from a few minutes to one hour, one hour to one week, several months, and several years, respectively. Among these categories, STLF has gained popularity due to the proliferation of internet of things (IoT) applications based on smart meter data, the increasing granularity of data for data-driven techniques, and the ease of establishing data relevance with conditional factors [97]. STLF serves as a key input for operational decisions in unit commitment and source coordination to minimise transmission costs [64]. Consequently, STLF plays a pivotal role in enhancing the efficiency of the power supply side,

ensuring grid safety, establishing common ground for complex pricing strategies in the energy market, and addressing challenges stemming from the increased integration of RESs [115,116]. Moreover, STLF serves as a point of reference for DR applications to adapt to price changes or instant regulations in the electricity market. Regulatory actions can be achieved through load curtailment and load shifting. However, repricing is typically determined through tariff adjustment procedures.

LF is also assessed from the perspective of economical electricity dispatching, as accurately estimating future demand is crucial for meeting demand while minimising costs. In this context, challenges arise in

Table 1 (continued).

[50]	2020	RNN-LSTM	To facilitate a comparative analysis, the study first discloses the input–output relationship of STLF data and offers a suitable model. Subsequently, it identifies the most appropriate combination of feature sets and an efficient encoding mechanism.
[21]	2020	LSTM, CNN, GRU and stacked autoencoder	To test the performance of competitive DL techniques by a real case study.
[51]	2020	Video pixel networks based one-dimensional CNNs with LSTM or GRU techniques	To obtain superior performance by apply point estimation instead of confidence interval estimation for the time series forecast.
[52]	2020	Sequence to sequence and a two dimensional CNNs	To obtain adequate prediction accuracy under poor information case.
[53]	2020	Multi-objective DBN with empirical mode decomposition	To introduce a LF model that offers superior accuracy and greater generalisation ability compared to similar competitive methods.
[54]	2020	Supervised ML forecasting models by Gaussian kernel regression with random feature expansion and nonparametric based k -NN	To introduce novel LD forecast models with enhanced predictive potential for speed and accuracy.
[55]	2020	Quantile regression CNN	To demonstrate the effectiveness of the proposed model in comparison to other competitive approaches.
[5]	2020	CNN for probabilistic LF	To obtain more reliable and sharper load probability distributions by probabilistic LF models.
[56]	2020	Hybrid forecast model including modified mutual information, factored conditional restricted BMN, and genetic wind driven optimisation techniques	To propose adaptive DL approach for LF.
[57]	2020	Cluster-based aggregate forecasting	To offer random clustering baselines to get higher accuracy in load prediction.
[58]	2020	Hybrid model of empirical mode decomposition and GRU	To enhance the forecasting performance, a hybrid procedure that incorporates the decomposing process instead of using LSTM directly is proposed.
[59]	2020	LSTM	To show the effectiveness of LSTM against the traditional networks.
[60]	2020	Blended decision tree, k -NN, feedforward NN, and deep feedforward NN	To consider various factors, including date, exact time, climate variables, and artificial predictive variables using the proposed mixed model.
[61]	2020	LSTM and deep feedforward NN	To capture both members' interdependencies by LSTM and climate factors on LF by deep feedforward NN.
[62]	2020	DL-based electricity demand forecasting model by sequence to sequence regression technique	To include input ranking feature selection analysis with RReliefF algorithm.
[63]	2020	LSTM	To propose priority based framework for economical utilisation of household devices.
[64]	2020	TCN	To present precocious accuracy in LF problem by extracting the auxiliary input from the residential consumption components by the proposed framework.
[65]	2020	LSTM-GRU	To develop time series forecasting model by some RNN types for the agriculture load.
[66]	2020	Combination of LSTM k-means clustering	To uncover similarities in consumption patterns that enable the forecasting model to effectively mitigate the overfitting issue.
[67]	2020	RNN	To propose an incentive based DR program together with DL-based forecasting model.
[68]	2021	Online adaptive RNN	To achieve higher accuracy than the stand-alone offline LSTM network and some online algorithms. They employ an adaptation mechanism that allows them to capture newly arriving data.
[69]	2021	Hybrid interval forecasting model combining k -NN optimised by genetic algorithm (GA), DBN and self-adaptive kernel density estimation techniques	To demonstrate the success of the proposed interval forecasting model in terms of accuracy and flexibility, without compromising the simplicity of the forecasting process.
[70]	2021	Trilinear deep residual network with self-adaptive dropout method based on hierarchical clustering and Gaussian noise.	To propose a robust model that mitigates the issues of vanishing and exploding gradients, along with addressing overfitting, while simultaneously achieving improved forecasting accuracy.

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terms of LF accuracy, price dynamics, the intermittent characteristics of energy renewables, and various operational constraints, as highlighted in studies like [121,122]. Achieving adequate accuracy in LF is an ongoing research focus, with efforts aimed at improving accuracy through factors such as climate dynamics, addressing uncertainties in demand, and employing risk assessment approaches for unexpected events.

Dispatching decisions are predominantly influenced by price fluctuations in the available supplied power. To ensure a fair and competitive energy market, reliability is a crucial parameter that can be better assessed through improved LF performance. Moreover, considering the increasing integration of renewables into the grid, these

sources produce intermittent and variable energy. Therefore, accurate LF prediction is vital for operators to balance unexpected deficits and ensure demand reliability.

Finally, operational constraints, including generator capacity, voltage-frequency stability, and transmission limits can also be incorporated into the economical dispatching of energy using LF concepts. These constraints have practical applications in optimising DR programs based on LF.

LF involves heavily varying non-linear dynamics due to the unpredictable nature of human-based activities or micro-climate variables. The methods used to address these challenges need to capture both short- and long-term dependencies, which are influenced by human behaviours, natural dynamics, and weather-related factors.

Table 1 (continued).

[71]	2021	The algorithms of concrete dropouts, deep ensembles, Bayesian NNs, deep Gaussian processes, and functional neural processes	To explore the probabilistic extensions and performance potential of DL algorithms.
[72]	2021	k -means CNN-LSTM forecast model with clustering approach	To obtain reliable energy consumption data for an academic building for DR application.
[73]	2021	Non-linear fully connected feed-forward ANN by autoencoder with localised stochastic sensitivity	To propose a DL model aimed at enhancing prediction accuracy and reliability by minimising the error, which is represented by the training error and stochastic sensitivity.
[74]	2021	Bidirectional LSTM based sequence to sequence regression approach	To evaluate the proposed model against other competitive techniques for both public holidays and normal days in terms of accuracy and the availability of limited data.
[75]	2021	Ensemble learning model using multi-modal multi-objective evolutionary algorithm and random vector functional link network based ensemble learning	To discover additional trade-off multimodal solutions by harnessing the mapping capabilities of the proposed ensemble learning approach in the context of STLTF problems.
[76]	2021	Asynchronous deep reinforcement learning (RL) based model with deterministic policy gradient	To address the high temporal correlation and convergence instability issues of STLTF using a deep RL model.
[77]	2021	Deep RNN	To increase the forecasting accuracy and performance under the uncertain model dynamics.
[78]	2021	CNN	To enhance the model's ability to capture non-linear relationships, a feature selection process is proposed.
[79]	2021	RNN, vanilla LSTM, stacked LSTM, bidirectional LSTM and GRU	To evaluate the performance of LF by comparing RNN, three variants of the LSTM model, and GRU.
[80]	2021	Deep RL	To consider using a pre-trained dataset instead of a random one when proposing LF results to optimise DR applications.
[81]	2021	Hybrid network consisted the layers of autoencoder LSTM, bidirectional LSTM, and stack of LSTM	To demonstrate the superiority of the proposed hybrid model when tested with the dataset collected from a residential home, as compared to previous similar studies.
[82]	2021	Comparative analysis with linear regression, tree-based regression, linear support vector machine (SVM), quadratic SVM, cubic SVM and RNN	To evaluate the performance of various ML and DL-based residential LF models.
[83]	2021	A prioritised experience replay automated RL	To provide a coupled approach with multi period forecasting and DR program.
[84]	2021	Past vector similarity	To forecast the load at the finer granularity by extracting the exact load patterns of the occupants regarding to their routine and socio-economic values.
[85]	2021	CNN with squeeze-and-excitation modules	To represent the strong relationship between climates variable and residential volatile LD by the proposed model.
[86]	2021	RNN with LSTM	To assess the predictive performance of the proposed model in comparison to other models that use the same dataset.
[87]	2021	Residential LF framework combined by k -means clustering algorithm and federated learning	To establish a cooperative training procedure by utilising fine-grained monitored consumption data.
[88]	2021	Separate use of LSTM and GRU	To show that the accuracy performance of STLTF better than the longer focused forecasting models.
[89]	2021	Deep forward NN by automated selecting the best Box-Jenkins models	To obtain higher accuracy than the shallow networks.
[90]	2021	CNN sequence to sequence model with an attention mechanism based on a multi-task learning method	To demonstrate the superior accuracy performance of the proposed model.
[91]	2022	Backward-eliminated exhaustive ensemble model for future selection method, and the LF techniques of k -NN, CNN, RNN and SVR.	To obtain higher accuracy by proposing backward-eliminated exhaustive approach for feature selection technique.
[92]	2022	LSTM by mix-up and transfer learning techniques	To propose reliable model by considering the lack of sufficient historical data on the consumption which reduces the accuracy.
[93]	2022	LSTM, federated stochastic gradient descent and federated averaging.	To train a single federated learning based model when dealing with multiple smart meters, eliminating the need to share local data.
[94]	2022	Ensemble model with LSTM, GRU, and TCN	To demonstrate that the proposed ensemble models achieve better performance than the traditional individual models.
[95]	2022	CNN based on wavelet and varying mode decomposition	To extract more detailed spectral and temporal information in order to enhance the forecasting performance, especially when exogenous data are not available.

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STLTF has been tackled using three primary methods: mathematical algorithms, single artificial intelligence algorithms, and combinatorial algorithms [100]. Mathematical algorithms encompass regression-based models, interpolations, serial methods, statistical techniques, and fuzzy-logic approaches. The majority of these models have been designed with a linear approach. However, linear or affine-linear methods exhibit poor performance when dealing with demand profiles that

exhibit random and volatile patterns. Non-linear models outperform linear approaches but may struggle to capture the volatility in demand patterns.

Intelligent algorithms, including ML methods, have been proposed, such as artificial neural networks (ANNs), RNNs, LSTMs, GRUs, support vector regression (SVR), BMN, decision and regression trees, random forest (RF), k -nearest neighbours (k -NNs), linear regression, Gaussian

Table 1 (continued).

[96]	2022	Federated learning model with ANN architecture	To address privacy and security requirements for residential LF through the smart meter data with dynamic power demand.
[97]	2022	Consecutive applications of STLTF network with a layer of GRUs and STLTF network constructed by stacking several TCNs	To enhance the DL-based elastic model while maintaining good performance under varying conditions, including changes in accommodation, temperature, humidity, and wind speed.
[98]	2022	Hybrid model including the CNN and an attention-based sequence to sequence network.	To improve the forecasting performance by capturing the long-term spatial and temporal features of the data.
[99]	2022	Two stage encoder-decoder architecture based on receptive field based dilated causal convolutional and bidirectional LSTM networks.	To increase the STLTF performance by encoder-decoder configuration.
[100]	2022	LSTM with back propagation NN and XGBoost	To find a moderate solution to the contradiction between forecasting accuracy and calculation speed.
[101]	2022	A dynamic ANN model motivated by meta-learning	To introduce a fine-tuning approach for predicting highly non-stationary points and implementing a robust forecasting procedure.
[102]	2022	Ensemble structure based on LSTM and XGBoost	To propose a more accurate and scalable model while mitigating some of the limitations of current approaches.
[103]	2022	Bidirectional LSTM	To propose seasonal segmentation to achieve relatively higher accuracy in the forecasting procedure by taking into consideration the seasonal factors specific to the dataset of the geographical territory.
[4]	2022	A self-adaptive DL model with particle swarm optimisation (PSO)	To improve the accuracy, robustness, repeatability and self-adaptive capability in load prediction.
[104]	2022	Ensemble model with XGBoost and light-gradient boosting machine (GBM), RF regression and stacking regressor	To analyse the correlation between various variables in the dataset and testing the model performance for most influential variables.
[105]	2022	A multi-channel bidirectional nested LSTM framework	To improve the prediction accuracy by following multiple sub-signals processing approach.
[106]	2022	XGBoost	To determine the occurrence range of peak load considering the load, weather and time factors.
[107]	2022	Autoencoder based LSTM	To propose a dual-channel structure in the encoder part for extracting different levels of time series data. Additionally, they suggest a three-channel output structure in the decoder part to enhance the model's representation ability.
[108]	2022	Hybrid model called as variational autoencoder bidirectional LSTM	To present the effectiveness of the proposed method over the classical models.
[109]	2022	Hybrid model with integrated GA bidirectional GRU	To present more stable and reliable model than the models developed by the classical methods.
[110]	2022	ML models of SVR, RF, XGBoost, light-GBM, adaptive boosting, bidirectional LSTM, GRU, and a DL regression model.	To specify best features and searching for nest ML model for predicting the hourly demand.
[111]	2022	Hybrid structure with empirical mode decomposition, one-dimensional CNN, TCN, a self-attention mechanism, and a LSTM	To propose hybrid model having more stable and accurate prediction for STLTF problem.
[112]	2022	ML approach with deep ANN and decision tree-based prediction	To show that the ML algorithms and regression analysis have adequate accuracy for LF.
[113]	2022	Hybrid structure with CNN, LSTM and MLP	To propose a solution that offers both adequate accuracy and robustness for LF problems.
[114]	2022	Joint structure with multi-feature fusion, self-attention mechanism, convolutional graph network	To obtain better prediction performance than some of the benchmark models.

processes ensemble, clustering, hybrid algorithms, k-means, fuzzy c-means algorithms, deep belief networks (DBNs), extreme learning machines, principal component regression, adaptive neuro-fuzzy inference systems, GBM, CNNs, and autoencoders. These ML-based models possess a robust mapping ability for STLTF, but they may be susceptible to increased forecasting errors caused by sudden deviations in seasonal or occupancy-related parameters [75].

To address these deficiencies, combinatorial approaches, categorised as hybrid or ensemble models, have been proposed. Hybrid models that split the task into two sub-tasks can reduce forecasting errors when dealing with aggregated loads but may fall short in capturing long-term dependencies in residential demand loads [98]. Additionally, ensemble forecasting strategies involve basic models combined through multiple learning algorithms. The output of each model is weighted according to the correlation degree to enhance accuracy. In the literature, both static and dynamic ensemble learning approaches have been proposed for electricity LF, utilising combinations of LSTM, GRU, and temporal convolutional networks (TCNs) techniques [94].

4. DL-based STLTF with different datasets

DL is a subfield of ML that utilises ANNs with multiple layers to learn and represent complex mappings between input and output

data [123,124]. The DL approach gained significant momentum shortly after the introduction of MLP and the development of back-propagation algorithms in the 1980s and 1990s [10]. DL techniques address the vanishing gradient problem more effectively compared to shallow models. Consequently, DL-based models exhibit superior capabilities in handling complex functions with high accuracy [6]. The DL perspective has also recently gained momentum due to the unprecedented availability of extensive datasets and advanced algorithms [10,123].

DL methods construct linear or non-linear functions to reduce the correlation between the input data and output response as closely as possible through multi-layer network models. Hence, the ML techniques mentioned in the previous two sections have been transformed into DL models by augmenting the mapping layers. Considering the LF problem, DL approaches require building complex networks which present superior advantages over classical techniques for multi-point scenarios in the profile. However, they also have disadvantages in terms of computational burden and restrictions to deterministic point forecasting [69]. Despite dealing with the pros and cons of DL approaches for LF, their immense capability to capture short- and long-term dependencies on input data and have tractable options for computational issues make them even more preferable than shallow learning models. The influential factors on electricity consumption can be classified

Table 2

Dataset specifications.

LT	Ref.	Origin	SR	Dataset
Residential load	[32,33,40,105]	UK	1 h	Domestic appliance-level electricity (UK-DALE) dataset
	[67,100]	Austin/USA	1 h	Pecan street inc dataport dataset
	[92]	California/USA	1 h	OpenEI-National renewable energy laboratory dataset
	[68,93]	London/UK	1 h	London hydro dataset
	[42,96]	London/UK	30 m	Low carbon London project dataset
	[31]	USA	1 h&1–5–30 m	UMass trace repository/Smart* datasets
	[95]	Morocco	1 h	Moroccan buildings' electricity consumption dataset
	[97]	France&Belgium	1&10 m	The Individual household electric power consumption and the appliances energy prediction (AEP) datasets
	[99]	Lahore/Pakistan	15 m	Lahore electric supply company dataset
	[43,58,77,86]	California/USA	1 m–1 h	UCI ML repository dataset (UCI public dataset)
	[34]	USA	1 m	UMass smart home dataset
	[98]	California/USA	1 h	Individual household electric power consumption dataset
	[74]	Pakistan	1 d	FESCO dataset
	[64]	Canada	1 m	AMPds2 dataset
	[41]	Ireland&Australia	30 m	The Irish commission for energy regulation and Australian SG city project datasets
	[81]	Paris/France	1 d	The individual household electric power consumption (IHEPC) Dataset
	[63]	Alaska/USA	1 h	Open energy data initiative (OEDI) dataset
	[87]	Sydney/Australian	1 h	Australian “SG, Smart City” customer trial dataset
	[66]	Ireland	30 m	Smart metering dataset on the electricity customer behaviour trials initiated by the commission for energy regulation in Ireland
	[84]	New South Wales/Australia and Eskilstuna/Sweden	1 h	Specific
	[82]	Tallinn/Estonia	1 m	Specific
	[117]	Tomsk/Russia	1 d	Specific
	[29,50]	Jiangsu/China	1 m-1 h-,	Specific
	[28]	New England-Ireland/UK	1 h	Specific
Operator load	[94]	Algeria	5 m	Bejaia concession of Algerian electricity and gas distribution company (SADEG) dataset
	[3,56]	USA	5 m	PJM electricity market dataset
	[51]	Istanbul/Turkey	1 h	CK Bogazici Elektrik dataset
	[101]	Belgium	1 h	Elia: Belgian transmission system operator dataset
	[44,53,75]	Tasmania-Queensland-New South Wales/Australia	30–60 m	Australian energy market operator (AEMO)
	[45]	Eastren China&Eastren Australia	15 m	Ausgrid dataset
	[47,62]	Delhi	1 h	Tata power Delhi distribution limited dataset
	[3,5,21,70,76,89,111]	New England/USA	1 h	Independent System Operator-New England(ISO-NE) dataset
	[55,85]	USA	1 h	Global energy forecasting competition (GEFCOM) 2014 dataset
	[76]	USA	1 h	Global energy forecasting competition (GEFCOM) 2017 dataset
	[73]	France, Germany, Romania, and Spain	1 h	ENTSO-E data portal
	[37]	Chandigarh/India	1 d	UT Chandigarh dataset
	[107]	Canada	1 h	Alberta electric system operator
	[27]	Macedonia	1 h	Electricity transmission system operator of Macedonia (MEPSO) dataset
	[109]	Bangladesh	30 m	Bangladesh power system network dataset
	[46]	Tianjin/China	15 m	China Tianjin power grid
	[86]	Maine/USA	1 h	Electricity data - U.S. energy information administration (EIA)
	[36]	Los Angeles-USA	15 m	Los Angeles department of water and power (LADWP) dataset
	[88]	France	1 h	Dataset on the daily electricity load of France
	[48]	Scotland	1 h	Dataset of distribution networks of Scotland
	[91]	Czech Republic	30 m	Czech Republic electricity load dataset
	[110]	Panama	1 h	Panama's power system dataset
	[112]	Ontario/Canada	1 h	Independent electricity system operator (IESO)
	[113]	Mayotte	30 m	Electricity of Mayotte (EDM) dataset
	[113]	Panama	1 h	Panama Electricity LF from Kaggle dataset
	[114]	China	1 h	Shandong Data open network in China - Grid bus load dataset
	[61,80]	Toronto-Ontario/Canada	1 h-5 m	Independent electricity system operator (IESO) dataset
	[108]	Tabriz/Iran	1 h	Distribution network of Tabriz city dataset
	[71]	Germany	1 h	Specific
	[59]	Tien Giang/Vietnam	1 d	Specific
	[118]	Zhejiang/China and New Zealand	1 h	Specific
	[106]	Shanghai/China	30 m	Specific
	[103]	Madhya Pradesh/India	15 m	Specific
	[102]	Spain	10 m	Specisific
	[69]	Australia	30 m	Specific
	[83]	China	1 h	Specific

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as date-related, atmospheric, and economic factors [29]. Moreover, specifying the precision of the prediction can be directly related to the

Table 2 (continued).

Other loads	[90]	China	1 h	Integrated energy microgrid dataset on industrial loads
	[65]	India	30 m	MERC Interim Report, AMR dataset (agriculture load)
	[4]	Bristol/England	1 h	Campus building dataset
	[119]	South Korea	1 h	Korea electric power corporation (KEPCO) dataset on industrial load
	[78]	Zurich/Switzerland	1 h	The building data genome project dataset on public buildings
	[30]	Japan	30 m	Power company dataset on industrial load
	[120]	China	30 m	Carbon culture platform dataset on public buildings
	[72]	Bombay/India	15 m	Laboratory SEL academic building dataset
	[52]	Shandong/China	1 d	Specific

feasible identifications of each decisive or descriptive variable in the load profile, as well as the learning type of the DL models. DL models have frequently been referenced in the literature for the LF problem. Table 1 reviews the studies on the DL-based STLF problem in terms of the proposed techniques and the main ideas of the corresponding studies. This table provides a broad perspective for prospective research on DL-based STLF problems. It summarises the related studies in the literature in terms of the connection between the adopted techniques and potential contributions to the literature. In this manner, the proposed techniques have been specified along with the main objectives of the study to clarify the potential of these studies. Hence, new research on the proposed solution techniques can be initiated easily, as well as potential comparative analyses may be dealt with.

Table 2 presents the specifications of the dataset from the paper referenced by the keywords of DL and STLF. In Table 2, LT, reference (Ref) studies, dataset origin, data SR and the specific dataset are given, respectively. LT defines the types of load such as residential, operator or aggregated, industrial, agriculture or public buildings loads. In this manner, operator data is logged as mixed types of data by utility companies in aggregated form. Besides, the origin shows the country in which the data is collected. SR is the granularity of the data at the corresponding study, and the time resolution may be various for the dataset origin. For SR, the time periods of minute, hour and day are abbreviated as m, h and d, respectively. Finally, the dataset specifies the type and specific source of the collected data. The dataset may be taken from data metering centres which are given in Table 2 or the data is obtained by the authors. For the second case, the dataset part of the figure is typed as specific.

5. Frequently referenced techniques for DL-based STLF problem

DL techniques in Table 1 can be viewed as evolved or extended versions of ANNs. Therefore, it can be said that DL-based models are multi-layered NNs, where input signals pass through multiple layers of neurons to generate an output. The distinctive feature of DL is the depth of the networks, signifying they consist of a significant number of hidden layers. These layers can extract progressively complex features from the input data, enabling the network to learn and classify patterns with higher accuracy. Moreover, DL techniques may include specific architectures optimised for processing images and sequential data, respectively.

Based on the information presented in Table 1, one can identify the most frequently referenced techniques as follows: RNNs, variants of RNNs such as LSTM and GRU, CNNs, and autoencoders. The frequency of occurrence of these techniques is calculated by considering the references reviewed in the manuscript. LSTM, CNN, RNN, GRU, and autoencoders have respective shares of 32%, 17%, 10%, 9%, and 8%.

These techniques were developed to address the limitations of previous approaches or to tackle different types of problems. Various factors are taken into account during the application process, making it impractical and unfair to provide a general comparative analysis. Nevertheless, these techniques can be compared based on key criteria, such as their ability to handle sequential data, modelling complexity, computational efficiency, and training time. The comparative results are presented in Table 3. Additionally, we offer a brief summary of these techniques in the following subsections. For more in-depth information, readers can refer to the Refs [125,126].

5.1. RNNs

RNN is a type of ANN designed for processing sequential data, such as speech, text, and time series [123,143]. Unlike traditional feedforward NNs, which process input data as a fixed-length vector, RNNs process input data as a sequence, where each time step in the sequence is connected to the previous time step. The basic idea behind an RNN is to use a hidden state vector to capture the information from previous time steps, which is then used to make predictions at each time step. The hidden state vector is updated based on the current input and the previous hidden state at each time step. This allows the network to capture the long-term dependencies in sequential data, making it well-suited for tasks such as language modelling, speech recognition, and time series forecasting.

In terms of structure, an RNN is composed of a series of interconnected neurons, where each neuron is connected to the neurons in the previous and next time steps. The weights of the connections between neurons are shared across all time steps, allowing the network to learn patterns in the data that persist over a long period of time. A typical RNN flowchart includes an input layer, one or more hidden layers, and an output layer, as illustrated in Fig. 2. The input layer takes the data for a given time step and passes it to the hidden layer. The hidden layer consists of a set of neurons, each with its own set of weights and biases. The hidden layer processes the input and generates an output, which is then fed back into the hidden layer as the hidden state for the next time step. This hidden state captures information from previous time steps, which is used to make predictions at each time step. The output layer takes the outputs from the hidden layer and generates the final prediction. The weights (P_1 , P_2 , and P_3) and biases of the network are learned through training, where the network is exposed to a series of inputs and their corresponding outputs, and the weights and biases are adjusted to minimise the error between the network's predictions and the actual outputs.

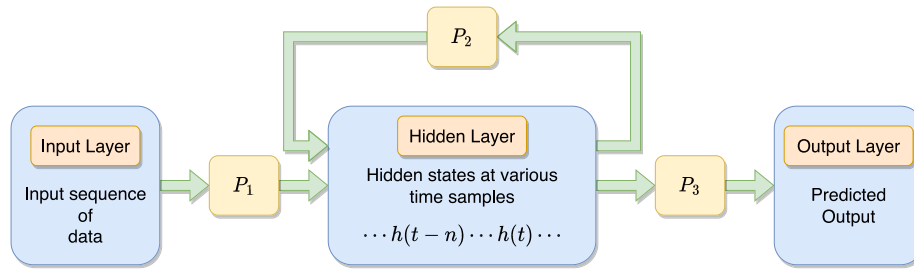
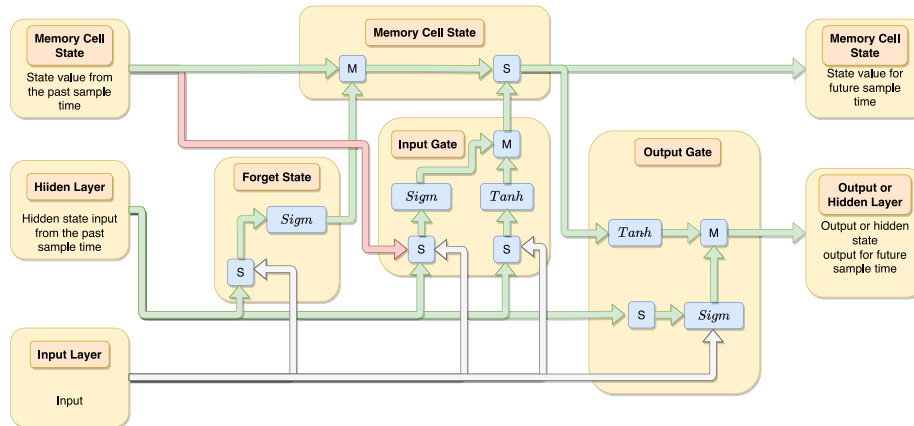
RNNs are well-suited for processing sequential data and have been utilised in LF to model the temporal dependencies in the data. Several variants of RNNs have been developed in recent years, including simple RNN, LSTM network, GRU network, bidirectional RNN, and CNN-LSTM network. Each of these variants has its own strengths and weaknesses, and the choice of which one to use will depend on the specific problem at hand and the type of data being employed. When considering the DL-based STLF problem, LSTM, GRU, and CNN-LSTM are frequently referenced.

In addressing the design decisions for RNNs, it is crucial to consider two key aspects: sequential processing and the vanishing gradient issue. Real-world data sources often exhibit temporal dependencies. RNNs capture these temporal patterns through sequential processing, achieved by employing recurrent connections with hidden states. Additionally, the vanishing gradient problem is closely linked to these recurrent connections within the model. As a result, weak recurrent connections can limit the network's capacity to capture long-term dependencies, leading to challenges associated with the vanishing gradient problem.

Table 3

Comparative analysis of advantages and disadvantages of RNN, LSTM, GRU, CNN, and autoencoders in terms of key factors.

Benchmarks	The pros and cons of the techniques
Handling sequential data	RNN is convenient for handling sequential data [127]. LSTM and GRU are suitable for sequential data, with LSTM being effective in mitigating the vanishing gradient problem [128]. CNN is primarily designed for spatial data, characterised by multidimensional arrays, rather than sequential data [129]. Autoencoders can also be utilised for processing sequential data arrays [130,131].
Modelling complexity	Simple RNNs are less complex than LSTM and GRU [132]. The LSTM concept includes three gates, while GRUs have fewer parameters since they merge the input and forget gates into the update gate [132]. Hence, it can be inferred that GRU is simpler than LSTM as it includes fewer gating structures. Moreover, CNNs and autoencoders have different architectures compared to the RNN family, which makes it challenging to compare them directly. CNN architecture is built with numerous layers, each serving specific functions [133]. Due to limitations in their architecture, CNNs may benefit from optimisation algorithms to mitigate overfitting or issues with local minima [133]. The complexity of CNN modelling depends on the number of layers and extensions. Autoencoders, on the other hand, are a special form of feedforward architecture with encoder, decoder, and latent space, and they can also be utilised in convolutional or recurrent forms [123]. The complexity of autoencoders can be specified by the number of latent features [134].
Computational efficiency	Computational efficiency depends heavily on some special factors of model size, data type, batch size, hardware etc. Considering the same conditions, GRUs are computationally more efficient than LSTM [135]. In a general manner, GRU gives better performance than LSTM for low-complex cases, and LSTM performs better in relatively higher-complex strings [136]. Also, RNN family is executable on embedded platforms [137]. For general comparative analysis, CNNs are computationally intensive [138,139]. Autoencoders have advantageous for computational point of view as they can deal large dataset by dimensionality reduction algorithms [131].
Training time	Due to the exploding and the vanishing gradient problem, training of simple modellings may be longer compared by LSTM which is proposed to tackle with the gradient problem. GRUs are faster than LSTMs [135]. Training time for CNN is depended on layer numbers, as well as the weights [140]. Besides, weight sharing feature reduces the number of trainable parameters, as well as it avoids the overfitting [139]. For autoencoders, data feeding through the layers may impose bottleneck [141]. Besides, the size of the training-set can be optimised [142].

**Fig. 2.** RNN block diagram.**Fig. 3.** Single LSTM cell block diagram.

5.2. LSTM networks

LSTM networks were first introduced in [144]. LSTM is an advanced variant of RNN that is specifically designed to overcome the vanishing gradient problem common in traditional RNNs. In traditional RNNs, gradients used to update the model parameters during training can become very small over time, making it difficult to train the model effectively [145]. LSTM networks can efficiently extract the underlying complicated temporal connections using gates and memory cells, illustrated in Fig. 3. In Fig. 3, *Sigm* and *Tanh* stand for the sigmoid and hyperbolic tangent functions, respectively. Additionally, consecutive *S* and *M* indicate summation and multiplication, respectively. LSTM

networks accomplish the temporal pattern connections by using a type of memory cell designed to keep information over a long period of time. The memory cell is surrounded by gates that control the flow of information into and out of the cell. Due to their remarkable ability to capture long-term dependencies in long sequences of data, they are used in a variety of applications, including natural language processing, speech recognition, and LF.

They use memory cells and gates to keep information over a long period and improve the model's ability to learn from long sequences of data. As seen from the block diagram of a single unit in Fig. 3, the LSTM network has typical components of an input layer, memory cell, input gate, forget gate, output gate, hidden state, and output layer.

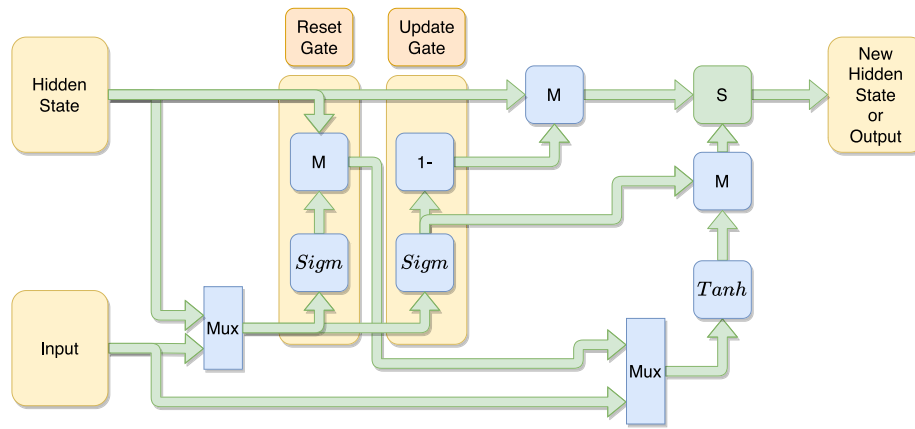


Fig. 4. GRU block diagram.

The input layer takes in the data the network will process, which is usually a sequence of values, such as time series data. The memory cell is surrounded by three gates: the input gate, the forget gate, and the output gate. The memory cell is the core component of the LSTM network, as it keeps the information from one time step to the next by carrying it over a long period of time.

The input gate controls the flow of information into the memory cell by determining what information should be added to the cell's state based on the current input and the previous cell state. Similarly, the forget gate controls the flow of information out of the memory cell by determining what information should be forgotten based on the current input and the previous cell state. The output gate is the third gate and controls the flow of information out of the memory cell and into the final output of the LSTM network. It determines what information should be output based on the current input and the previous cell state. In this regard, the hidden state is also the output of the memory cell after it has been processed by the input, forget, and output gates. The hidden state is used as the input for the next time step in the sequence.

Finally, the output layer takes the hidden state from the memory cell and produces the final output of the LSTM network, which can be used for prediction or classification tasks.

For the design decision of LSTM sequential processing, memory cells, and complexity should be considered. LSTMs are mainly designed for forecasting problems with sequential data. Hence, the order of the input is important to capture the temporal patterns. LSTMs include memory cells gating mechanisms and the design decision should be made by the capacity of the information capturing over long sequences due to the requirements to avoid the vanishing gradient problem. Also, LSTMs have relatively higher capacity to involve the long-term dependencies. But this capacity increasing costs to higher computational burden.

5.3. GRU networks

GRU network is another variation of RNNs that has been frequently used in LF to capture both short-term and long-term dependencies in the data [146]. As seen from Fig. 4, GRU consists of the update and reset gates. The update gate decides how much information from the previous sample time should be passed on to the current step, while the reset gate specifies how much of the previous information should be forgotten. These gates are implemented using sigmoid activation functions, which produce values between 0 and 1. The output of the GRU at each sample is determined by a combination of the previous state and the input at the current time. The state information is stored in the hidden units and updated at each time sample based on the input and the gate values. In this regard, the critical advantage of GRUs over traditional RNNs is that they are able to effectively balance the trade-off between retaining old information and incorporating new information,

allowing them to model long-term dependencies in sequential data more effectively. This has made GRUs popular choice for a wide range of sequence modelling tasks, as well as LF problems.

GRU and LSTM are two types of RNNs designed to address the problem of vanishing gradients, which can occur when training RNNs on long sequences. GRU networks have been introduced as a simplified version of LSTM networks. They have two gates to control the flow of information through the network instead of three gates in LSTMs. Also, GRU networks use a single memory unit to store information, while LSTMs use two memory units. Hence, they require relatively fewer parameters to be tuned and less training time compared to LSTMs.

GRUs are simpler alternatives to LSTMs. They incorporate only update-reset gates and have a more straightforward construction. This simplified form emerges as the primary design decision factor for GRUs, with the aim of balancing forecasting performance and computational burden for shorter sequenced forecasting problems. Furthermore, when compared to LSTM cells, GRUs have less complex memory cells, making them a suitable option.

5.4. CNNs

CNNs are a type of deep NN commonly employed for image recognition and computer vision tasks [147,148]. CNNs are designed to handle time series data and have been used to capture local patterns within time series data. A block diagram of a CNN is provided in Fig. 5. As observed in the diagram, a CNN consists of a series of convolution layers that extract features from the input image, followed by one or more fully connected layers that make predictions based on the extracted features. The input image undergoes convolution with a set of filters to generate a set of output feature maps. Each filter is a small matrix of weights slid over the image, computing a dot product at each position to create a single value in the output feature map. These filters are typically learned through back-propagation during training, and the values in the feature maps represent learned visual features, such as edges or corners, which are useful for identifying objects in the image. After each convolution layer, a non-linear activation function, such as rectified linear unit, is applied to the output feature maps to introduce non-linearity into the network. The feature maps are then typically down-sampled using pooling layers, reducing the spatial size of the features and enhancing the network's computational efficiency. Finally, one or more fully connected layers are added to the end of the network to make predictions based on the learned features. These layers are similar to those in a traditional NN and may use techniques such as dropout or batch normalisation to improve generalisation and prevent overfitting.

CNNs have significant differences from GRU or LSTM networks. CNNs use convolutional layers to extract features from input data, while GRU or LSTM networks use recurrent connections to maintain

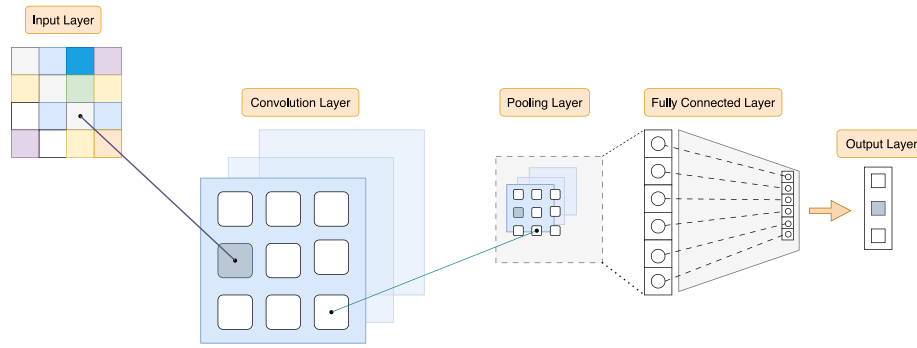


Fig. 5. CNN block diagram.

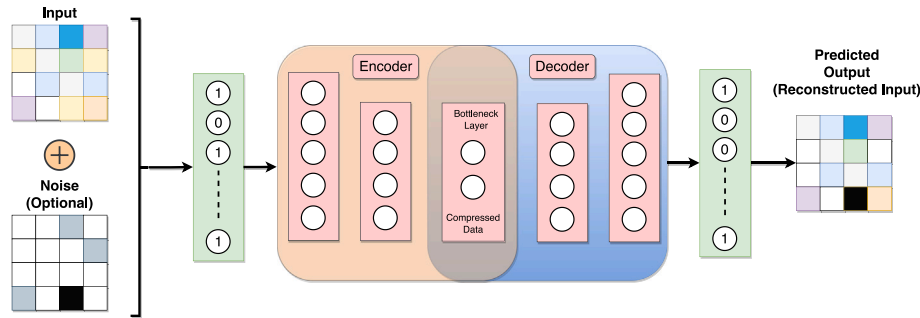


Fig. 6. Autoencoders diagram.

the memory of the input. CNNs process data in a spatially local manner by applying filters to small sub-regions of the input data. In contrast, GRU operates on data in a temporally local manner by using recurrent connections to maintain a memory of previous inputs over time. Therefore, CNN networks are well suited for processing visual data such as images and video, while GRU or LSTM networks are used for sequence data processing tasks.

When considering the design decision factors of CNNs, it is essential to take into account the roles of convolutional layers, translation invariance, and pooling layers. Convolutional layers serve to detect partial patterns within the input spatial data, enabling the network to learn features ranging from low-level local details to high-level dimensional details. The concept of translation invariance means that CNNs recognise features regardless of the object's position in the data. This simplifies the learning process by focusing on local features instead of requiring extensive feature engineering. To reduce the spatial dimension and facilitate down-sampling, CNNs incorporate pooling layers, which aid in managing computational complexity.

5.5. Autoencoders

Autoencoders are one of the most important ML models. They transform input data into a more compact and informative representation in the form of a vector set, which can be utilised for downstream tasks such as classification, clustering, or forecasting [149]. Encoding involves mapping raw input data to a lower-dimensional space by capturing the most notable features of the input. This process is achieved through a series of non-linear transformations that reconstruct the input data into a sequence of hidden representations. These depiction relations are learned from data using gradient descent and backpropagation algorithms. Autoencoders are often designed with an unsupervised learning approach where the goal is to learn a low-dimensional representation of the input data without any explicit supervision or labels. As shown in Fig. 6, autoencoders include basically two blocks which are the encoder and the decoder. The encoder network takes an input data sample and maps it to a lower-dimensional latent representation.

The size of this representation is typically much smaller than the input size, which allows the network to learn a compressed form of the input data. Then, the encoder network can be composed of one or more layers of neurons and can be implemented using a variety of NN architectures, such as fully connected layers or convolutional layers. However, the decoder network takes the compressed inherent representation produced by the encoder network and maps it back to a reconstructed output that is the same size as the input. So, it can be said that the decoder is typically a mirror image of the encoder in terms of the number of layers and the architecture type. Besides, the connections and weights are in reverse form to each other. Considering the training process, the objective of the autoencoders is to minimise the difference between the original input and the predicted output. This is typically done using a function defined by the metrics of mean squared error or cross-entropy loss. Autoencoders can be trained in an unsupervised manner. Hence, they can learn to extract useful features from input data without requiring explicit labels. Once trained, the encoder network can be used to extract features from new data samples, which can be especially used for an LF problem. Autoencoders can be efficiently used for unsupervised learning tasks, such as dimensionality reduction, feature extraction, anomaly detection etc. Besides, CNN, GRU or LSTM networks are primarily used for supervised learning tasks, such as classification, regression, and sequence modelling etc. Briefly, it can be said that autoencoders do not require labelled data for training which differentiates them from other competitive network models.

Considering the performance of each technique, autoencoders can be effective in learning useful representations of high-dimensional data. However, they may not perform well such as other network architectures on specific supervised learning tasks, especially those involving sequential or image data, where RNNs, GRUs, LSTMs, and CNNs have shown superior performance. For LF problems, autoencoders can be preferred for LF models with feature extraction and data pre-processing to improve the accuracy of forecasting. Also, they can be trained using an unsupervised learning approach, which provides immense help to model performance as this form of learning does not require labelled data. In this regard, in cases where limited training data is available, autoencoders can be designed with transfer learning mode, which enables

them to be pre-trained on a related task and then fine-tuned on the LF problem. Moreover, autoencoders are relatively successful in anomaly detection and predicting unusual events, such as power outages or equipment failures. However, autoencoders can suffer from overfitting when the size of the input data is large or the network architecture is complex. This can lead to poor generalisation performance on new data. Also, training autoencoders can be computationally expensive, particularly for large datasets or complex network architectures. Additionally, the design of autoencoders does not have a precise algorithm to determine the optimal architecture for a given task, which can be time-consuming and required expertise.

When considering design decisions for autoencoders, several key points should be taken into account, including the architecture, the bottleneck layer, and the use of variational autoencoders. Autoencoders consist of two internal networks that work in an integrated manner, allowing them to effectively learn features. The encoder network maps input data to a compressed latent image, and the decoder network reconstructs the data from this image. This structure results in the designing of a bottleneck layer in the middle of the model, which serves to reduce the dimension of the latent representation. Consequently, important features of the network can be captured in a more compact representation at this layer. Furthermore, variational autoencoders introduce a probabilistic element to the model, enabling the generation of new data samples for purposes such as data generation and anomaly detection.

6. Online LF models

Classical NN approaches utilise offline learning and miss the chance to learn from newly arriving data. They handle all data in the training procedure, and the resulting model serves to forecast future demands without adapting the inner parameters [68]. On the other hand, online ML-based approaches have the potential to embrace variations in the data stream and adapt to the model by reiterating the procedure with available data [68]. Considering the DL-based STLF concept, one of the future direction is developing online learning type of NN LF models. Online LF techniques are utilised to predict future electricity or energy demand in real-time or near real-time. Online measuring of continuously generated datasets enable precise LF which paves the way for the efficient power distribution programs to compete with the imbalances and energy shortage scenarios [150].

The basic online-type LF techniques include time series analysis, ML with statistical models, hybrid models, online learning algorithms, online data sources, probabilistic forecasting, and deep RL [150]. When searching for these techniques within the context of STLF, the literature is not extensive. Therefore, the STLF perspective has been broadened to encompass general horizon LF. Online-based LF techniques are reviewed to provide an analysis of the advantages and disadvantages of these techniques and to project their applicability to the LF concept.

In this context, online-type LF studies are presented in Table 4 in terms of the basic concepts and key findings. This table clarifies the fundamental models used to address various types of LF problems and briefly highlights the favourable results achieved.

When considering Table 4, certain issues can be specifically evaluated. For example, DL algorithms such as convolution or feed-forward NNs calculate the output based solely on the current input, without considering the temporal patterns of the data [68]. However, RNNs introduce forward mapping models based on both current and historical inputs. The concept of continuous learning and adapting to new patterns is studied in the context of online learning RNN [68]. This model has the capacity for continuous learning from newly arriving data, adapting itself by updating RNN weights to accommodate new patterns or changes in data type. Its performance has been compared to other online and offline algorithms. The efficient integration of the online learning concept into LF models is discussed, and an application of an online ensemble-type model is presented [151].

In the context of real-time data-driven models, a sliding-window online algorithm based on deep networks has been proposed [152]. The computational ability under volatile online parameters has been tested through an extensive long-term experimental procedure. This perspective is also relevant in an IoT environment. An online-type implementation of LF architecture has been introduced in Refs. [153, 154]. In this approach, issues related to data collection efficiency, caused by the inefficiencies in data meters, are addressed through online and transfer learning techniques using adaptive-based models with time-series data [155]. Furthermore, learning performance has been enhanced through a deep hybrid model featuring a CNN-LSTM network [156]. Probabilistic LF is also discussed within the context of adaptive online learning, taking into consideration changing consumption patterns and model parameters [157].

7. Uncertainty-aware LF models

LF is a process for estimating future demand based on recent data in national (deregulated) or liberalised energy markets. The load data obtained from those markets are uncertain and non-linear in nature, necessitating the use of DL methodologies. DL-based LF helps provide more robust solutions within the energy management framework in terms of cost, efficiency, and reliability [167]. Rigorous STLF is essential to recognise the absolute deficit in the balance of energy supply and demand. Considering the concept of liberalised power grids, rigorous STLF can also be used as an argument for a price forecasting model to make optimal decisions for system operators, energy suppliers, and end-users [27].

Network-based estimation procedures may experience decreased performance due to uncertainty in real power consumption, primarily because of incomplete metering infrastructure deployment. Additionally, inherent issues in NNs, such as overfitting, vanishing gradients, and exploding gradients, can gradually diminish the robustness of forecasting models. As a result, DL models with classical or customised NN techniques have been utilised in the literature to enhance performance in dealing with uncertainties.

Besides the focus on the DL-based STLF concept in this research, the literature is not rich in this area. Therefore, the search scope is expanded to include ML-based robust LF models in order to gather innovative ideas for improving the robustness of STLF. In this context, Table 5 is prepared to present the fundamental concepts and key findings related to uncertainty considerations in each of the referenced papers. Considering Table 5, studies related to uncertainty in the LF problem encompass both deterministic and probabilistic models. Probabilistic models are primarily referenced for representing uncertain dynamics, providing essential background information for robust LF models. Additionally, studies combining online optimisation procedures and statistical approaches offer a promising foundation for robust LF models. This approach not only allows for iterative parameter updates but also incorporates representations of random variables at specified confidence levels. On the other hand, this concept should be further developed to introduce new patterns that better capture the volatile nature of STLF and to streamline the complexity of the training process.

In Table 5, various single or hybrid ML techniques, including LSTM, CNN, RNN, RL, and GRU, are referenced in both probabilistic and deterministic robust LF modelling. Furthermore, many papers explore specific LF models supported by concepts such as parameter adaptation, reward incentive mechanisms, clustering, and wavelet processing. Optimisation algorithms like PSO, GA, and k-NN are employed for fine-tuning the model parameters. These key findings on the representation of uncertainty in model or exogenous parameters offer valuable insights for future research. Probabilistic LF approaches hold significant potential for representing parametric uncertainty, as exemplified in modelling peak load uncertainty in terms of occurrence

Table 4

Online type LF studies in terms of basic concept and key findings.

Ref.	Year	Basic concept	Key findings on online LF
[157]	2014	Autonomous kernel based online ANN model and its application to building cooling load prediction	Autonomous kernel-based NN model is proposed for noisy data regression prediction without requiring external update of the model parameters.
[158]	2015	RF model for STLF	Providing design scheme of online type RF techniques considering the day-ahead learning procedure for long range forecast.
[159]	2018	Kernel extreme learning machine and Cholesky decomposition for STLF	Taking into account the parameter updating of the growing dataset to overcome both of the short-term irregularity in power load and variation in weather conditions.
[150]	2018	Online SVR model for STLF	Presenting design details of online SVR and giving discussions on the pros and cons of the proposed method with the tree-based ensemble methods.
[156]	2020	Hybrid DL scheduling algorithm for STLF online learning system with CNN-LSTM model	Enhancing and accelerating the learning performance in a multi-client auto-metering dataset. Presenting the design of the learning system for edge-cloud computing.
[153]	2020	In an IoT environment, supervised ML algorithms are considered for online STLF, and the Gaussian process regression ML algorithm emerges as the best-fitting LF algorithm for the provided setup	An online LF architecture is proposed for use in an IoT-based online home energy management system, along with a basic circuitry design for real-time data logging onto the cloud.
[160]	2021	Ensemble STLF method based on passive aggressive regression algorithm	Answers the question “How can we efficiently integrate the online learning concept into the existing LF models to obtain better accuracy?” Developing an online ensemble model for LF applications.
[68]	2021	Online Adaptive RNNs for STLF	Building a foundational learner model using RNNs to capture temporal dependencies, alongside a tuning module to adapt the NN's hyperparameters to newly arriving patterns. Comparing the accuracy performance of the proposed model with online models such as MLP, linear regression, passive-aggressive algorithm, online bagging, and k -NN.
[152]	2021	A sliding-window type online learning algorithm with deep echo-state network for STLF	An online learning scheme is implemented to enhance edge computation capabilities and capture dynamic changes in LD. The results of long-term experiments demonstrate promising reductions in computational burden and improvements in accuracy for real-world residential load data.
[161]	2021	DL-based sequential encoder-stacked decoder model for STLF	Presenting an online training method for sequence data based on transfer learning. Comparative analysis is provided between the hidden Markov model and RNNs.
[162]	2021	Adaptive online ensemble learning with RNN and ARIMA for STLF	Combining approach with ARIMA and RNN for STLF in drift concept. Online adaptive RNN to capture time dependencies in the dataset. Considering the overall LF error, online ARIMA-RNN ensemble performs better than standalone online adaptive RNN or ARIMA.
[154]	2021	Hidden Markov model based online learning algorithm for adaptive probabilistic LF	Adapting LF model parameters to the changes in the consumption patterns and load uncertainties. Providing online learning techniques for probabilistic LF model to update the model parameters. Presenting efficient implementation of online learning for adaptive probabilistic LF.
[163]	2022	Two-stage online learning-based STLF model called as for RAnger-Based online learning approach	To further improve the performance of the online learning-based stacking ensemble model, an online learning approach is proposed for better capturing the rapid changes or variability in building energy consumption trends.
[164]	2022	A physical-based multiple linear regression model for cooling LF	Online type model promises the better characteristics in terms of small training samples, generalisation ability, and interpret ability. Together utilisation of online training&calibration improves the performance by mean percentage absolute error compared to the offline models.
[165]	2022	Hybrid tree-based ensemble learning model including GBM, extreme GB, Cubist, and RF for daily peak LF	Utilising the Boruta algorithm to select all admissible features to build online learning model, and online model is trained on these variables using time-series cross-validation.
[151]	2022	Time-delay NN model for STLF	Time-delay embedding theory is utilised for the LF to confirm interpretability of the model. Proposing hour-ahead LF by online training and testing approach. Proposed algorithm is suitable for IoT-based networks with smaller dataset.
[155]	2023	Adaptive DL-based STLF framework by integrating transformer and domain knowledge	Considering the data collection performance due to inadequate deployment of the data meters, as well as data privacy issues, it is aimed to offer adaptive based model for time-series forecasting.
[166]	2023	Online-offline DL method for residential STLF	Proposing a simple structure for effective online training, as well as remarkable performance to learn the temporal dynamics. Following the feasibility analysis of applying online and offline learning scheme with historical and real-time dataset for LF.

and magnitude [168] and load probability distribution [5]. Additionally, optimisation techniques such as GA and PSO are proposed for hyper-parameter optimisation and assessing prediction uncertainty levels during training [69,169]. The optimisation procedures also enhance the convergence rate of the LF model [69].

Considering Table 5, various DL-based robust LF models have been developed using techniques such as adaptive and reward incentive methods, ensemble probabilistic learning, CNN, RNN, GRU, and clustering. In this context, the application of deep RNN techniques aims to enhance LF accuracy and performance under uncertain modelling dynamics [170]. Additionally, deep RL, which combines the non-linear fitting capabilities of DL with the decision-making abilities of RL, has provided effective solutions to optimisation problems involving both certain and uncertain dynamics. For example, a robust framework using an asynchronous deep RL based model is presented in [76]. This research has conducted multiple comparative experiments to achieve robust LF results.

Moreover, probabilistic forecasting techniques have the potential to represent the high level of uncertainty in real-time LD with great accuracy. Regarding the probabilistic concept, the utilisation of CNN, combined with load range discretisation for LF, is proposed [5]. This method results in more precise load probability estimates, which are beneficial for decision-making processes in DR applications and provide a significant advantage in feature extraction.

Furthermore, in pursuit of higher levels of stability and reliability against the uncertain and volatile nature of residential load data, a probabilistic LF model with a multitask Bayesian DL approach is presented [41]. Another robust probabilistic baseline estimation framework is provided for quantifying consumer responsiveness to estimate pricing inputs [42]. Clustering approaches have also been employed in STLF to analyse time-series data and identify both certain and uncertain patterns based on different time scales.

8. Practical justification of DL-based STLF studies

Recently, DL-based STLF studies achieve significance due to their practical justifications in improving the forecasting accuracy, model efficiency, and reliability of electricity grid operations [11]. Besides, LF should be combined with short-term DR programs that manipulate actual energy demands. DR programs aim to reduce electricity demand and maintain a balance between supply and demand by incentivising the shifting of energy usage to off-peak periods. These programs save significant energy and relieve strain on the power grid by aligning with LD. Also, they can leverage LF to encourage consumers to adjust their electricity usage in real-time [179]. In this regard, the practical and effective implementation of DR programs coupled with LF promise the benefits of data integration, customer engagement for participation, smart automation concept and fair competitive market regulations.

Considering utility-based planning and management strategies for electricity generation and distribution, accurate LF has a significant impact on the effectiveness of DR programs. Therefore, LF models should incorporate a wide range of data sources, including historical energy usage patterns, weather forecasts, occupancy data, and other specific factors relevant to the particular problem. Moreover, these models should possess the theoretical and practical capabilities necessary to provide near-real-time predictions. In this context, one study introduced a deep RL based forecasting model to enhance the accuracy of predicting RES outputs and loads, with the DR program designed around the predicted load [83]. Another study integrated a DL-based approach for daily energy demand prediction into a home energy management system, allowing for a reduction in overall appliance energy costs through linear programming [34]. Another study presented a deep RL approach for designing DR procedures, considering tariff structures based on LF, to provide a more economically efficient energy utilisation strategy for end-users [80]. In a separate effort, a DR management system was designed using a time-of-use-based approach, along with an

LSTM-based LF model to predict household appliance energy consumption [63]. Furthermore, an incentive-based DR program, incorporating customised DL and RL, was proposed, considering LF procedures, electricity prices, and photovoltaic power output [67]. Experimental results demonstrated a 17% reduction in peak electricity demand, mitigating supply-demand imbalances through the proposed approach. Highlighting the importance of accurate LF for the design of more robust and cost-effective DR strategies, a combined DL-based LSTM LF model was proposed to address multi-timescale electricity consumption behaviour for household residents [100]. Dealing with precise LF and economic analysis, differentiating between over-forecasting and under-forecasting, have been examined using both hybrid and single LF methods [61]. Results indicate that hybrid deep NN-based models outperform other methods, emphasising their significant contribution to the success of DR programs.

Dynamic pricing or tariff-based approaches are also associated with DL-based STLF problems. This concept also plays a crucial role in DR or general energy management strategies. Hence, the literature also contains examples of pricing notions in the related problem. To emphasise the important role of STLF in power system planning, a multi-scale CNN-LSTM hybrid NN STLF model is presented to reveal the correlation between electricity price and load [180]. The relationship between price and LF is also mentioned by proposing multiple-inputs, multiple-outputs deep RNNs models [3]. The dynamic time-of-use tariff procedure is dealt with by a clustering-based residential load estimation process [42]. In this study, the aim is to discuss some basic requirements for DR applications with time-varying pricing schedules.

Finally, we will discuss shallow NN models for the LF problem, providing valuable insights for DR applications. These models can be converted into DL format by appropriately adjusting the hidden layers. For example, the optimal program for energy demand is realised for a single household by using ML algorithms to forecast power generation and load aggregation [181]. In this model, the output of the forecasting model is used as input for the optimal scheduling procedure.

Another DR study is conducted using forecasting LD data obtained through a new prosumer microgrid criterion with hybrid ML techniques [182]. Unlike other DR studies that consider pre-defined load patterns without using any LF methods, this study shows that the adopted approach leads to more realistic results. By utilising the IoT concept, compact studies with LF and DR have also been realised. Decision trees and k -NN algorithms are proposed for STLF and price forecasting in a DR program [183]. The non-intrusive load monitoring method is proposed to identify the active LD, as well as adapt them to the appliance usage patterns [184]. This study helps predict household loads in fewer minutes, which is of significant practical interest for DR applications. A multi-agent RL-based model with a DR application is provided by using current load and all-day load prediction to shift higher demand to the period of low demand [185]. A framework for customer baseline load estimation for the day of a DR program is realised through load pattern forecasting [186].

LF is executed using unsupervised learning-based data mining approaches with self-organising maps and k -means clustering. An efficient energy management system was developed using hundreds of digital passive infrared occupancy sensors to detect data anomalies [187]. This setup utilises an LSTM network for day-ahead energy consumption forecasting, particularly for potential DR applications. Real-time training of RNNs using a parallel processing ANN approach is formulated for STLF [188]. A comparative analysis is proposed to show the suitability of GA and PSO to train the NN model for real-time LF [189]. Real-time anomaly detection for STLF is presented as a dynamic regression model [190]. Moreover, a unified ML-based training data generator concept with a look-back optimiser, which simultaneously performs electrical LF and unsupervised anomaly detection in real-time, is provided [191]. Real-time short-term residential load consumption problems are handled by RNN-LSTM techniques to offer a DR scheme to reduce peak LD [192].

Table 5

Uncertainty-aware LF studies in terms of basic concept and key findings.

Ref.	Year	Basic concept	Key findings on uncertainty
[169]	2014	PSO-based lower-upper bound estimation method for LF	A PSO-based method is proposed for lower-upper bound estimation in LF, as well as a PSO-based probabilistic optimisation formulation to tune the NN parameters, which allows for the formation of prediction intervals for uncertainty quantification.
[171]	2018	LF model by combination of generalised extreme learning machine, wavelet processing and bootstrapping techniques	The uncertainty formulations for the LF model and data noise are provided using a probabilistic approach.
[172]	2018	Hybrid model with empirical mode decomposition, DBN and ensemble technique for short-term cooling LF	In the synthesis of a robust model, statistical analysis is conducted to identify imperfections in the model, and necessary modifications are introduced to the formulation using ensemble techniques.
[173]	2018	Quantile regression forest enhanced by recursive feature elimination model for LF	The merits of probabilistic forecasting under uncertain parameters are highlighted. The level of quantile forecasting is considered in the model to include the uncertain patterns in the load.
[42]	2019	DL-based clustering method for STLF	A probabilistic estimation framework is proposed to quantify the electricity consumption responsiveness at the household level by considering the confidence level demand uncertainty.
[174]	2019	Probabilistic model for peak LF	The focus is on addressing uncertainties in different weather conditions and peak load using a probabilistic LF approach. The uncertainty occurred in the magnitude and occurrence pattern of the peak load is considered by the probabilistic model.
[175]	2019	DL-based ensemble probabilistic model	The probabilistic LF framework provides customer classification and multitask representation learning approach to make better uncertainty quantification of individual customers. The framework includes combination of deep ensemble learning and multitask representation to improve the performance for data analysis.
[176]	2020	STLF model by RL technique with Q-learning based dynamic model selection	Parameterised uncertainty of the deterministic LF, and optimising the uncertainty indicator by minimising the pinball loss with GA.
[5]	2020	DL-based CNN network for probabilistic STLF	The study provides solution for adaption of the probabilistic LF concept to the CNNs which considers the uncertain cases in the decision variables. A long range discretisation method is presented to define load probability distributions which enables the segmentation of the load using in training procedure.
[41]	2020	Multitask Bayesian DL network for residential probabilistic LF	Multitask learning probabilistic LF is proposed to quantify simultaneously the mutual uncertainties over specific end-user groups, as well as separating the uncertainty pattern for each group.
[170]	2020	DL-based hybrid model with RNNs and clustering techniques for STLF	Detailed analysis of future selection and hyper-parameter fine-tuning mechanisms for the LF accuracy and robustness are given. Proposed model performs well to the variational cases of building type, building operation, seasonal patterns etc.
[76]	2021	Asynchronous deep reinforcement deterministic policy gradient model with adaptive early forecasting method and reward incentive mechanism for STLF	Robustness analysis for the proposed model is tested by experimental studies. Dataset is divided into two cases which include different training sets and test sets to verify the robustness of the proposed model.
[177]	2021	Theory-guided deep-learning LF	Presenting model has the capacity of training procedure by random disturbance originated by weather and load inaccuracies.
[178]	2021	The deep mixture network for residential STLF model with CNN, GRU, and fully-connected neural layers	Providing a deep mixture model to capture inherent intermittent uncertainty of the residential load profiles. Uncertainty awareness performance of the model is obtained for single household load, as well as aggregated loads.
[69]	2021	DBN STLF model by non-dominated sorting GA and k -NN	k -NN algorithm is optimised by non-dominated sorting GA, and compared to the other LF models, remarkable improvements are obtained in convergence speed and robustness. Proposed model performs credible robust results that are insensitive to the sample size and train-to-test ratio.
[168]	2022	End-to-end DL framework of dual-stage attention based quantile LSTM network for STLF	Pinball loss is adopted as the loss function to better capture load uncertainties. The proposed model promises robust performance against changing variable conditions by adaptively increasing the correlation level of features and reducing the impact of irrelevant variables.

9. Discussions, further suggestions and limitations

To enhance the feasibility, reliability, and robustness of DL-based STLF solutions, it is essential to focus on performance indicators such as sensitivity, uncertainty bands, and dataset uniformity. Additionally, extending the forecasting problem to include renewables managed by DR applications can introduce innovation by utilising real-time

data for feedback mechanisms. This approach establishes a closed-loop model, potentially accelerating convergence toward predefined success criteria. Furthermore, when considering internal parameter optimisation within DL models, the incorporation of evolutionary algorithms can lead to more accurate and robust forecasts. In light of these considerations, the following suggestions are proposed.

9.1. Sensitivity in DL models

Balancing sensitivity in DL models is crucial for addressing STLF challenges through feature selection, data smoothing, and model complexity considerations. In DL models, sensitivity refers to the responsiveness and flexibility of predictions when faced with variations in input data or parameters. Highly sensitive models may produce unreliable predictions in response to minor deviations in the input data. To address the STLF problem, various strategies and techniques can be employed to determine the optimal level of sensitivity. For instance, one approach involves the meticulous selection of input features that are less susceptible to noise or data fluctuations. Instead of relying solely on static data, incorporating factors such as temperature or occupancy trends within daily or even hourly averages can be considered. Furthermore, features can be normalised or scaled to emphasise relevant patterns. In addition, data-smoothing techniques like moving, exponential, or kernel smoothing can be introduced to reduce noise in the data, thereby making the model more sensitive. Although certain approaches, such as capturing both short- and long-term trends through hybrid models, combining multiple DL-based LF models into an ensemble framework, and including uncertainty estimation filters and extreme data detection mechanisms, may introduce complexity, they prove valuable in regulating sensitivity. Finally, certain key stages in the training process of DL model-based applications should be noted. Implementing cross-validation to identify and fine-tune the model makes predictions less sensitive to fluctuations in load data. On the other hand, the careful tuning of hyperparameters, as well as the judicious determination of the number of hidden layers, strikes a fine balance between model complexity and sensitivity.

9.2. Handling uncertainty in DL models

Addressing uncertainty in DL models through probabilistic forecasting, online feedback loops, and ensemble techniques enhances accuracy and robustness for STLF. The uncertainty in DL models can arise from inherent randomness or variability in the observed data and a lack of knowledge about certain modelling dynamics. Attempting to achieve complete modelling or eliminate data fluctuations is inconclusive. Therefore, the aim is to develop models that can handle uncertainty by considering parameters with changing trends to provide more accurate predictions. To accomplish this, the proposal includes an online feedback prediction loop that continuously monitors forecasting performance to update the model's parameters. This approach can also eliminate the influence arising from data logging external variables. The uncertainty-aware framework incorporates specific techniques, such as probabilistic LF, which provides a range of possible values, and quantifying uncertainties using NNs, hybrid or ensemble modelling. DL approaches gain robustness through probabilistic forecasting, as they provide a range of values rather than a single point estimate. Therefore, it is possible to integrate the probable distribution of values into the model's mapping mechanism. By using Bayesian NNs, the weights are treated as random variables with probability distributions, and prediction intervals can be extracted at specified confidence levels. Furthermore, hybrid and ensemble models can combine LF methods and leverage the strengths of each component to provide more accurate and robust predictions. Therefore, the techniques proposed here for achieving a more certain DL-based STLF model can be applied in future studies, as they capture distinct aspects of LD characteristics.

9.3. Innovating STLF with multi-agent systems and IoT

Expanding the capabilities of STLF through the integration of multi-agent systems and the IoT for real-time DR applications promises to revolutionise LF models and ensure heightened accuracy and coordination. Building upon the discussion in Section 8, the primary focus is on real-time implementations of DR applications for short-term energy

management. Furthermore, DL-based LF models can take on new, innovative forms by incorporating valuable theoretical and practical approaches, as well as groundbreaking ideas. The scope of this concept can be broadened by harnessing the power of multi-agent systems and the IoT.

The multi-agent approach is centred around systems in which multiple autonomous agents interact with each other and their environment. On the other hand, IoT pertains to the interconnection of physical objects, devices, and systems that collect and exchange data for autonomous operation. A fundamental distinction between multi-agent systems and the IoT concept lies in the fact that "IoT focuses on connecting physical devices for data sensing, while multi-agent systems focus on distributed decision-making among autonomous agents".

IoT and multi-agent systems significantly enhance the capabilities of LF models in terms of accuracy and reliability. Sensing and metering IoT devices provide real-time data on consumption, grid balance conditions, unexpected events, and equipment monitoring. Furthermore, multi-agent systems enable better coordination and utilisation of resources, facilitate load management actions (such as load shifting or shedding), decentralise decision-making, and ensure secure communication among the agents

9.4. Dataset uniformity for enhanced analysis

The performance of data-driven multi-layer models hinges on dataset uniformity and necessitates decoupling mutual effects among various LTs to design efficient DR applications. In the literature, the majority of datasets are obtained from operator-based loads, which encompass a mixed form of energy demand patterns. These patterns include various end-user types, such as residential, industrial, EV charging stations, and public buildings. Furthermore, some shared components of these LTs, like heating-ventilation-air conditioning or private charging units, significantly contribute to the total LD. Therefore, it is essential to decouple the mutual effects of these factors from each LT to enable feasible analysis. This approach allows for the ideal design of DR applications.

This situation raises the question of how to separate the basic LTs and distinguish the significant common sub-classes of loads. The answer to this question requires expanding the dataset's scope to include factors such as energy demand, occupancy patterns for different regional family types, social preference surveys, weather-based dynamics, and more. With such a comprehensive dataset, forecasting models can establish extensive mapping networks between input and output characteristics, enabling the implementation of versatile DR-based programs.

9.5. Integrating renewables and evolutionary algorithms

Combining DR with renewables and harnessing evolutionary algorithms promises to reduce load variability and enhance the accuracy and robustness of the STLF models. The sudden time-based variability in LD can be mitigated by utilising RESs in conjunction with DR applications. DR programs can influence end-users' usage habits by offering profitable incentives. As a result, the volatility of demand can be maintained at a reasonable level. Therefore, incorporating DR applications and renewables into the feedback dynamics of the LF problem reduces the magnitude of the challenge.

Evolutionary algorithms also improve the performance of STLF models in terms of accuracy and robustness. They can optimise future subsets of the search range, as well as various combinations of hyperparameters using feature selection criteria based on mutual information and forecasting scores. Evolutionary algorithms have an immense capacity to overcome trapping on a local minima. They provide candidate solutions in the vicinity of local minima to search for the global solution. When considering GAs, this process can be primarily managed by mutation operators that introduce random changes into the

candidate solutions. Therefore, it is proposed as a future direction to use GAs with multiple mutation operators [24]. In each iteration, a specific mutation operator is chosen and applied to the optimal hyperparameter search problem. This increases the randomness level of the solution, directly enhancing the convergence rate of the parameters to the global optimal points.

9.6. Empowering energy decision-makers with accurate STLF insights

Accurate STLF provides critical insights for informed decision-making in the energy sector, leading to optimised energy operations, enhanced grid stability, increased profitability, and reduced operational costs. Considering the valuable insights and benefits of LF, the players of the sector can be informed for making the right decisions on the operation procedures for some key factors. Those factors can be listed as optimised energy production and consumption, enhanced grid stability and more profitable trading and reducing the operational costs. Those factors can be subjected to the above-mentioned discussions for technical issues, as well as they can lead to the sector through the data drive decision making environment.

10. Conclusions

In this review, recent studies on STLF using DL were compiled. These studies were categorised based on their proposed techniques, primary objectives, and the datasets they employed. Succinct explanations were provided for frequently referenced techniques, and a comprehensive analysis of research related to DL-based STLF was conducted. This survey encompassed online and robust models, along with multi-objective investigations that integrated this approach with DR applications.

In conclusion, DL-based STLF has emerged as a popular research direction in the search for more accurate and high-performance solutions for LF. Improved LF can directly contribute to the economic welfare of the SG concept. Obtaining more reliable and faster STLF solutions can lead to the widespread adoption of DR applications in grid operations. Therefore, this review facilitates individuals seeking an understanding of the background of DL-based STLF problems and furnishes them with a succinct comparative analysis of the merits and drawbacks of the proposed techniques.

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Declaration of competing interest

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